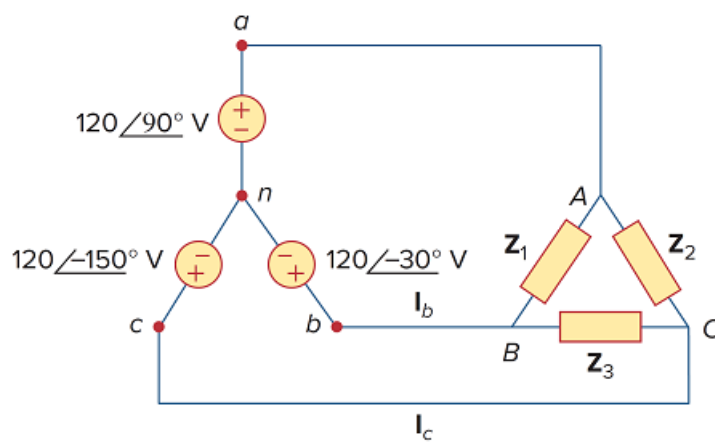


Problem

Consider the wye-delta system shown in Fig. 12.60. Let $\mathbf{Z}_1 = 100 \, \Omega$, $\mathbf{Z}_2 = j100 \, \Omega$, and $\mathbf{Z}_3 = -j100 \, \Omega$. Determine the phase currents, $\mathbf{I}_{AB}$, $\mathbf{I}_{BC}$, and $\mathbf{I}_{CA}$, and the line currents, $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, and $\mathbf{I}_{cC}$.

Figure 12.60



Step-by-step solution

Step 1 of 10

Refer to Figure 12.60 from the textbook.

Consider the current through $\mathbf{Z}_1$ be $\mathbf{I}_{AB}$, the current through $\mathbf{Z}_2$ be $\mathbf{I}_{CA}$ and the current through $\mathbf{Z}_3$ be $\mathbf{I}_{BC}$.

Consider the line currents as $\mathbf{I}_{aA}$, $\mathbf{I}_{bB}$, $\mathbf{I}_{cC}$.

Step 2 of 10

Apply the Kirchhoff's Voltage Law in the loop $aABbn$.

$$\mathbf{I}_{AB}\mathbf{Z}_1 + 120\angle -30^\circ - 120\angle 90^\circ = 0$$

$$\mathbf{I}_{AB}\mathbf{Z}_1 = 120\angle 90^\circ - 120\angle -30^\circ$$

$$\mathbf{I}_{AB} = \frac{120\angle 90^\circ - 120\angle -30^\circ}{\mathbf{Z}_1}$$

Step 3 of 10

Substitute $100 \, \Omega$ for $\mathbf{Z}_1$.

$$\mathbf{I}_{AB} = \frac{120\angle 90^\circ - 120\angle -30^\circ}{100}$$

$$= \frac{-103.92 + j180}{100}$$

$$= -1.039 + j1.8$$

$$= 2.078\angle 120^\circ$$

Therefore, the phase current, $\mathbf{I}_{AB}$ is $\boxed{2.078\angle 120^\circ \text{ A}}$.

Step 4 of 10

Apply the Kirchhoff's Voltage Law in the loop $cnbBC$.

$$\mathbf{I}_{BC}\mathbf{Z}_3 - 120\angle -30^\circ + 120\angle -150^\circ = 0$$

$$\mathbf{I}_{BC}\mathbf{Z}_3 = 120\angle -30^\circ - 120\angle -150^\circ$$

$$\mathbf{I}_{BC} = \frac{120\angle -30^\circ - 120\angle -150^\circ}{\mathbf{Z}_3}$$

Step 5 of 10

Substitute $-j100\ \Omega$ for $\mathbf{Z}_1$.

$$\mathbf{I}_{BC} = \frac{120\angle -30^\circ - 120\angle -150^\circ}{(-j100)}$$

$$= j\left(\frac{207.846}{100}\right)$$

$$= j2.078$$

$$= 2.078\angle 90^\circ$$

Therefore, the phase current, $\mathbf{I}_{BC}$ is $\boxed{2.078\angle 90^\circ\text{ A}}$.

Step 6 of 10

Apply the Kirchhoff's Voltage Law in the loop ABC .

$$\mathbf{I}_{AB}\mathbf{Z}_1 + \mathbf{I}_{CA}\mathbf{Z}_2 + \mathbf{I}_{BC}\mathbf{Z}_3 = 0$$

Substitute $2.078\angle 120^\circ\text{ A}$ for $\mathbf{I}_{AB}$, $2.078\angle 90^\circ\text{ A}$ for $\mathbf{I}_{BC}$, $100\ \Omega$ for $\mathbf{Z}_1$, $-j100\ \Omega$ for $\mathbf{Z}_1$ and $j100\ \Omega$ for $\mathbf{Z}_2$.

$$(2.078\angle 120^\circ)(100) + (2.078\angle 90^\circ)(-j100) + \mathbf{I}_{CA}(j100) = 0$$

$$207.8\angle 120^\circ + (2.078\angle 90^\circ)(100\angle -90^\circ) + \mathbf{I}_{CA}(j100) = 0$$

$$207.8\angle 120^\circ + 207.8\angle 0^\circ + \mathbf{I}_{CA}(j100) = 0$$

Step 7 of 10

Thus,

$$\mathbf{I}_{CA} = -\left(\frac{207.8\angle 0^\circ + 207.8\angle 120^\circ}{j100}\right)$$

$$= -\left(\frac{-103.9 + j179.96}{j100}\right)$$

$$= -1.799 + j1.039$$

$$= 2.078\angle 150^\circ$$

Therefore, the phase current, $\mathbf{I}_{CA}$ is $\boxed{2.078\angle 150^\circ\text{ A}}$.

Step 8 of 10

Apply the Kirchhoff's Current Law at node A .

$$\mathbf{I}_{aA} - \mathbf{I}_{AB} + \mathbf{I}_{CA} = 0$$

$$\mathbf{I}_{aA} = \mathbf{I}_{AB} - \mathbf{I}_{CA}$$

Substitute $2.078\angle 120^\circ$ A for $\mathbf{I}_{AB}$ and $2.078\angle 150^\circ$ A for $\mathbf{I}_{CA}$.

$$\begin{aligned}\mathbf{I}_{aA} &= 2.078\angle 120^\circ - 2.078\angle 150^\circ \\ &= -1.039 + j1.799 + 1.799 - j1.039 \\ &= 0.761 + j0.761 \\ &= 1.0756\angle 45^\circ\end{aligned}$$

Therefore, the line current, $\mathbf{I}_{aA}$ is $1.0756\angle 45^\circ$ A.

Step 9 of 10

Apply the Kirchhoff's Current Law at node B .

$$\mathbf{I}_{bB} + \mathbf{I}_{AB} - \mathbf{I}_{BC} = 0$$

$$\mathbf{I}_{bB} = \mathbf{I}_{BC} - \mathbf{I}_{AB}$$

Substitute $2.078\angle 120^\circ$ A for $\mathbf{I}_{AB}$ and $2.078\angle 90^\circ$ A for $\mathbf{I}_{BC}$.

$$\begin{aligned}\mathbf{I}_{aA} &= 2.078\angle 90^\circ - 2.078\angle 120^\circ \\ &= j2.078 - (-1.039 + j1.799) \\ &= 1.039 + j0.278 \\ &= 1.0756\angle 15^\circ\end{aligned}$$

Therefore, the line current, $\mathbf{I}_{aA}$ is $1.0756\angle 15^\circ$ A.

Step 10 of 10

Apply the Kirchhoff's Current Law at node C .

$$\mathbf{I}_{cC} + \mathbf{I}_{BC} - \mathbf{I}_{CA} = 0$$

$$\mathbf{I}_{cC} = \mathbf{I}_{CA} - \mathbf{I}_{BC}$$

Substitute $2.078\angle 150^\circ$ A for $\mathbf{I}_{CA}$ and $2.078\angle 90^\circ$ A for $\mathbf{I}_{BC}$.

$$\begin{aligned}\mathbf{I}_{aA} &= 2.078\angle 150^\circ - 2.078\angle 90^\circ \\ &= -1.799 + j1.039 - j2.078 \\ &= -1.799 - j1.039 \\ &= 2.078\angle -150^\circ\end{aligned}$$

Therefore, the line current, $\mathbf{I}_{aA}$ is $2.078\angle -150^\circ$ A.